

The Ephemeris

September 2023

Volume 34 Number 03- The Official Publication of the San Jose Astronomical Association



September 2023 - December 2023 Events

Please check our website for up to date information.

Board & General Meetings

Saturday 9/2 9/30, 10/28, 12/2

Board Meetings: 6 -7:30pm

General Meetings: 8-9:30pm

Solar Sunday (2-4pm)

Sunday 9/10, 10/8, 11/5, 12/3

In-Town Star Party

Friday 9/8, 10/6, 11/3, 12/15

RCDO Starry Nights Star Party

Saturday 9/9, 10/7, 11/4, 12/17

Imaging SIG Mtg (Online)

Tuesday 9/12, 10/17, 11/21, 12/12

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Pelican Nebula
Photo by SJAA Member: Hy Murveit
June 23, 2023
Location: Home

“I Would Rather See Starlight than Streetlights”

Credit: Marianne Damon

Which one of us would NOT want this bumper sticker? I saw it when I was renewing my membership to The Astronomical League. About ten years ago, prior to becoming very active with SJAA, I was hunting for astronomy information online. A frequent pop-up as a go-to for visual observers was the AL. I looked into it and decided the programs were well-designed and offered to be done at my own pace. I decided it would give structure to my desire to increase my observing skills. It was a good place to start and from the looks of it, a good place to continue learning and stretching observational abilities over a lifetime.

Many astronomy groups join in toto as a Member Society. Our association does not have a person to help facilitate this, so the option is for individuals to join as members-at-large (adults \$40, youth \$20). With your membership and subsequent renewals, you are entitled to a choice of one Astronomical League observing guide for free. Over the years with my renewals, I've collected Beginner's Messier Objects, Universe Sampler, A Pathway to the Stars, Handbook for Astronomy Educators, Lunar Observations, and Carbon Stars. And there are a few more to go. You can also begin/renew subscriptions for Sky & Telescope, Astronomy and Star Date all in one place. I do this and find it very convenient. Their online store also sells RASC handbooks and calendars (those often sell out quickly), as well as other observing handbooks and merchandise. I just purchased two Northern Hemisphere sky bandanas for my grandchildren.

The AL administers over 30 Observing Programs, numerous National and Youth Awards, and supports Outreach and Dark Sky Preservation efforts. Their quarterly publication is the Reflector. Each Observing Program has a coordinator and a list of rules and requirements; upon completion a pin is awarded. I must confess that due to my dilettante nature, my name does not yet appear on any of their lists, but I've completed a handful of programs to my satisfaction. A fond memory is of our late, beloved Marilyn Perry who acquired a curio-cabinet full of AL pins; she was my inspiration for starting the Lunar Observing Program.

You can find a listing of all of their programs at their website. Here is the link to the listing of their observing programs: [Alphabetical Listing – Offerings from the Observing Program Division of the AL - Astronomical League \(astroleague.org\)](#) A sampling includes programs for binoculars, ancient and modern constellations, lunar observing, solar observing, variable star observing, and sketching, just to name a few. Common to each program is the requirement to list seeing and transparency as part of your observation log. The programs address skills all the way from Sky Puppies for young beginners under ten (membership not required) to Master Observer and Master Imager.

The AL has officially been around for seventy-seven years, but it was in the planning stages for far longer (several choke points were the Great Depression and a World War). It was founded as a group of amateur astronomers for amateur astronomers. Currently there are 300 member societies and 23,000 members. Here is the link to their main page: [Home - Astronomical League \(astroleague.org\)](#).

Happy Hours

Credit: Marianne Damon

Over the years I've been impressed with the time and effort that our members put in to make our association run smoothly. We volunteer our time because we enjoy learning and are passionate about sharing our knowledge with other members and the public. We are paid not in dollars but in satisfaction. However, sometimes a dollar amount attached to statistics can provide an extra "wow" factor, and a deeper appreciation of the value of our efforts.

I decided to make an attempt to tally up our volunteer hours. I sent out a request for information to our officers, board members and program leads. Thank you to all of you who responded. For those that didn't, I used my best judgement from my knowledge of your programs to make an estimate. Knowing that our events are often weather related and that there can be great variations in the go/no-go status, the final numbers will have a fairly wide margin of error. And for activities that can't be individually measured, I've come up with a set amount of "reasonable" time. For instance, In Town Star Parties and Starry Nights have an average of ten volunteers; I've estimated their round-trip drive time at an hour and their set-up/tear-down time at an hour in addition to the public viewing time. Seasonal and quarterly programs have been figured and then converted to hours per month.

MEMBERS ONLY PROGRAMS:

Board Meetings: Let's start with officers and board members. Many of them have other roles, but this encompasses preparation, scheduling, communicating, driving time, setting up tables/chairs/electronics and attendance at the Board meetings. Some members, such as our President and Secretary, report more hours than others. We have four officers and nine directors, but since three of the officers are also board members, the total required attendance is ten persons. Total estimated hours: 40 per month.

Member Observing: We have members who obtain permits for our viewing areas and are docents for these programs. They set up schedules and put out calls for viewing at Mendoza Ranch, Rancho Canada del Oro, Little Uvas, and several other trial sites. There are nights for observers and for imagers. The docents arrive early, open gates, monitor arrivals, and stay to close. Total estimated hours: 25 per month of docent time.

Loaner Program: With coordination and 2-4 people helping, estimate is about 10 hours per month.

Quick START: About three sessions per year, 4-5 hours of training; 4 members involved. 5 hours per month.

Donations: Arrangements, travel time, meeting & load up and storage. 4 hours per month.

Library: Currently on hold due to vacancy; estimate of 4 hours per month.

Ephemeris: Content and Production editors, articles written by members, review by Board members. 4 hours per month.

Webmaster: 2 hours per month. **Social Media:** 2 hours per month **Membership:** 2 hours per month **Volunteer Coordinator:** 2 hours per month

Special Task Forces: 24" Dob, 5 members, 5 hours per month; **Code of Conduct/Minors Policy,** 6 members, 8 hours per month; Meeting A-V Electronics, 3-4 members, 4 hours per month. City of San Jose Compliance, 3 members, 4 hours per month.

PUBLIC PROGRAMS:

Monthly Speaker/Meeting: Obtain speaker, hall set-up, shop & set up refreshments, audio set-up, cleanup and trash removal. 15 hours per month.

In Town Star Parties: 50 hours per month. **Starry Nights:** 50 hours per month. **Solar Observing:** 45 hours per month. **Astronomy 101/Beginner Astronomy Class:** 8 hours per month.

Astronomy Beginner's Forum: A new online program! Supported by 3-4 members each session. 6 hours per month.

Binocular Star Gazing: 3-4 per summer. 6 hours per month. **Pinnacles Star Parties:** 3-4 per summer. 10 hours per month. **School Star Parties:** 10 hours per month

Imaging SIG & Lectures: This very popular program has an email list of 288! The hours include organizing the speakers and also the prep time when one of our members is the speaker. 10 hours per month.

Our monthly hours total for members only programs is 121 and our total for public programs is 210, for a grand total of 331 hours per month. Since our current membership is 367, this indicates that our members are out there and active! Now for the application of the final part of this report, the dollar amount that these hours represent. The current hourly rate for non-profit organizations in California (for 2022 per Independent Sector) is \$37.32. Our total monthly volunteer contribution to our association and our community is...wait for it...\$12,353!

Citizen science: building the light curve of supernova SN 2023ixf

Credit: Francesco Meschia

On the night of May 19, the Japanese amateur astronomer Koichi Itagaki noticed a new star in one of the spiral arms of the M101 galaxy. The new star was soon determined to be a Type II supernova, and given the designation name SN 2023ixf.

Just a few hours later, our own Arvind Kalyan shared the news on the SJAA Astroimaging list, and I quickly prepared a sequence in NINA to image M101, collecting the first data of what would become my 80-day run of photometry on this supernova.

A Type II supernova is the explosion that follows the collapse of the core of a star, of mass between 8 and 40 times that of the Sun, that has exceeded the Chandrasekhar limit and has catastrophically turned itself into a neutron star. A powerful shock wave tears the star apart, and the release of energy makes the star briefly brighter than its entire host galaxy, so that it can be seen from distances of tens or hundreds of millions of light years.

Supernovae are interesting for astronomers because they're a window into stellar evolution, and their observational data can be used to validate and refine models. Also, their peak intrinsic brightness is predictable, and so they can be used as "standard candles" to calculate their distance based on their apparent brightness. This is especially true for Type I supernovae, but the light curve of a Type II supernova is still useful for calibrating the distance scale.

Measuring the brightness, or magnitude, of an astronomical object has, for a long time, been the province of expert visual observers, who trained their eyes and brains to quantitatively express the difference in luminosity between stars that they saw at the eyepiece. This is the way that organizations like AAVSO collected thousands of light curves, mostly from amateur astronomers, for over a century.

But today, many of us have electronic cameras, which can measure how much light falls onto each of millions of photosites on their sensors. While I greatly enjoy using them for the challenge of imaging deep-sky objects, these cameras are also instruments that can be used to produce quantitative measurements of position and brightness, and of how they change over time.

The method I used to measure the brightness of SN 2023ixf is known as "aperture photometry." It is a measurement technique that uses the data from digital imagery. The digital values in each pixel are converted into electron counts, and the counts of all the pixels found within a certain aperture that includes the target to measure are added together. The result represents the total number of electrons produced by the light from both the target and the background. Then, an estimate is made of the numbers of electrons that likely have come from the background, for instance by counting the electrons from pixels in a ring ("annulus") around the aperture that only or mostly contains background signal. The background contribution is subtracted from the total, to obtain a number that is a proxy for the luminous flux collected from the target.

If similar flux measurements are collected for a set of catalog stars of fixed brightness, of which the corresponding stellar magnitudes are known, then the magnitude of the target object can be calculated, knowing the logarithmic law that correlates magnitudes with flux. This method is known as differential photometry.

Of course, there's more to differential photometry than just this. For instance, the number of electrons released in the sensor by the light from a target depends on how the incoming light is filtered before getting to the sensor. A broadband filter will let more radiation flux pass through than a narrower one. A blue filter will yield a lower flux count from a carbon star than a red filter.

Star catalogs typically list several magnitude values for a star, originating from different filters with well-known spectral characteristics, like the set of Johnson-Cousins UBVRI filters or the set of Sloan ugrIZ filters. Strictly speaking, differential photometry should only be done by comparing apples to apples: V catalog magnitudes with fluxes obtained through the same V filter, and so on.

When this condition is not met, for instance when a different filter is used to capture the images used for aperture photometry, there will be some systematic error introduced in the measurement, which will depend on the spectral composition of the light from the target, and the spectral characteristics of the two filters.

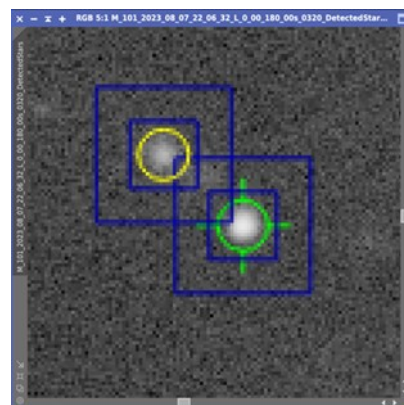


Figure 1 – Aperture photometry. The green and yellow circles represent the photometric apertures, and the blue squares denote the regions ("annuli") used to estimate the background contribution.

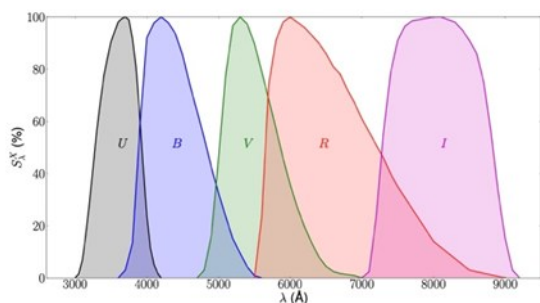


Figure 2 – Typical transmission curves for the Johnson-Cousins UBVRI filter set.

Source: Coelho, Calvão, Reis, “Standardization of type Ia supernovae.”

Many of us imagers don't have ready access to calibrated photometric filters, but we do have either a set of LRGB filters or the filters embedded in the sensor's Bayer matrix. When I started doing my photometry campaign on SN 2023ixf, I chose to use my L (luminance) filter, because it gave me the best signal-to-noise ratio thanks to its broad passband. Also, I decided to use V magnitudes from AAVSO's Photometric All-Sky Survey (APASS) as a reference, so that my measurements would end up being converted to V magnitudes. I had previously run a simulation to estimate what error would be introduced by the difference between the two filters, for stars with blackbody temperatures from 3000K to 8000K. The results led me to expect that my instrumental error would remain within 0.05 magnitudes even if the supernova ran the entire temperature gamut during the observation period. A G filter from an RGB set would have been a good choice too, giving a spectral response closer to that of the Johnson V filter, but at the price of lower signal-to-noise ratio.

Every clear night from late May to early August I ran a short imaging sequence, right at the beginning of astronomical night, collecting 5 exposures of 120 seconds initially, or 180 seconds in the later phases. I calibrated each image with a master dark and a master flat, and I used PixInsight's Aperture Photometry script on them.

This script does aperture photometry by counting the electrons within a circular or square aperture centered on all the catalog stars found in the image, and on one or more objects to be measured that are specified by the user. It can estimate the background contribution by various methods, such as using a square photometric annulus, or a wavelet transform on the image to determine its background, or the well-known Automatic Background Extraction process (which is the one I chose to use). It then produces a text file containing, for each of the catalog stars and user-defined objects, the catalog magnitude, the measured background-subtracted flux and its ratio to the background noise among other information.

This file doesn't give, by itself, the magnitude of the target objects: it only contains the *aperture* photometry information. The *differential* photometry part is left to the user. In my case, I chose to write a Python program that would process the file and compute a logarithmic fit between the measured fluxes and catalog magnitudes. Once the equation describing the fit is determined, so is the magnitude of the target object.

The program completes the measurement with an uncertainty figure, calculated via Gaussian error propagation from the uncertainty on the target object's flux and on the fit coefficient.

After the first month of photometry, I decided to share a preliminary light curve with a message posted on the SJAA Observers and Astroimaging mailing lists. In reply to my message, Stuart Heggie suggested I could share my data with the professional astronomers' community, but I didn't know how to do it. Stuart then kindly put me in contact with Dr. Malhar Kendurkar of the Prince George Observatory in Canada, whose research interest lies in the field of transient phenomena, such as supernovae.

Dr. Kendurkar was very open to the contribution, and very patient with me. He asked me to share my raw data and calibration frames first, and ran them through his own photometry pipeline. He then told me that my measurements were within 0.02 magnitudes of what he obtained at the observatory, and suggested that we could publish the entire light curve as an AstroNote on the Transient Name Server of the International Astronomical Union (<https://www.wis-tns.org/astronotes/astronote/2023-212>).

This entire photometry run was a very interesting and rewarding experience for me. It opened my eyes to the fact that we can do actual citizen science with our equipment, just by taking a different look at the data we collect as part of our imaging projects, and by applying simple but rigorous methods. Having a professional astronomer review my raw data, data reduction pipeline, and results, gave me confidence that my techniques were viable and that I can use them again in future projects. And, at some point, I may even see my data quoted in a scientific paper.

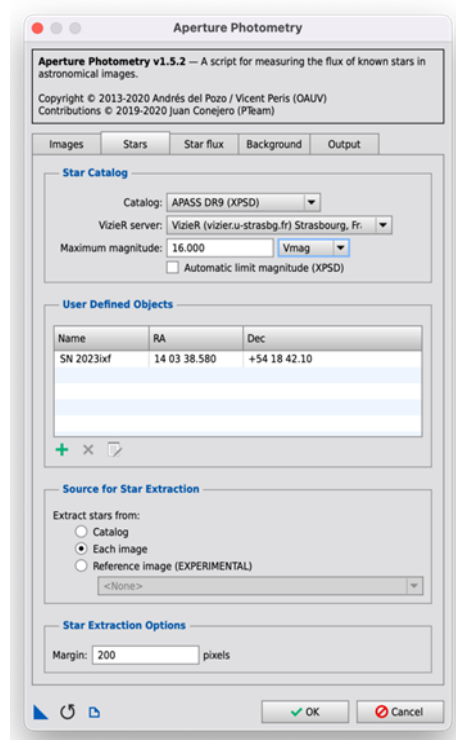


Figure 3 – The Aperture Photometry script dialog, showing the catalog selection and the coordinates of the objects to measure.

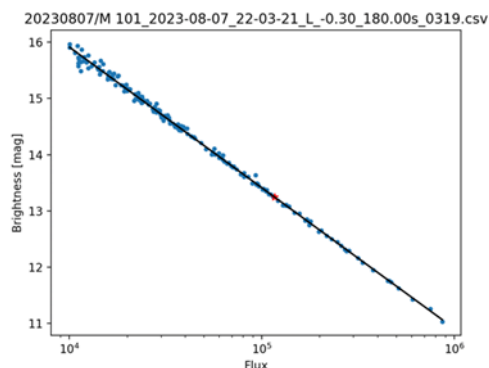


Figure 4 – An example of logarithmic fit between brightness expressed as magnitude and flux. The blue dots represent catalog stars, the red marker represents the supernova.

Photometry and astrometry (e.g., measuring the positions of asteroids or comets) are certainly two fields in which amateurs can give useful contributions. NASA, for instance, is looking for amateurs willing to collect and contribute photometry data for the confirmation and study of exoplanets (<https://exoplanets.nasa.gov/exoplanet-watch/about-exoplanet-watch/overview/>). Dr. Don Bruns, who in 2017 replicated the Eddington Experiment during the Great American Solar Eclipse, is looking for imagers to join his new project to experimentally measure the deflection of starlight around Jupiter this coming October (see <https://skyandtelescope.org/get-involved/pro-am-collaboration/how-to-measure-jupiters-deflection-of-starlight/>).

There's plenty of opportunity for us to put our equipment to new, creative, and scientific uses. It's up to us to seize it.

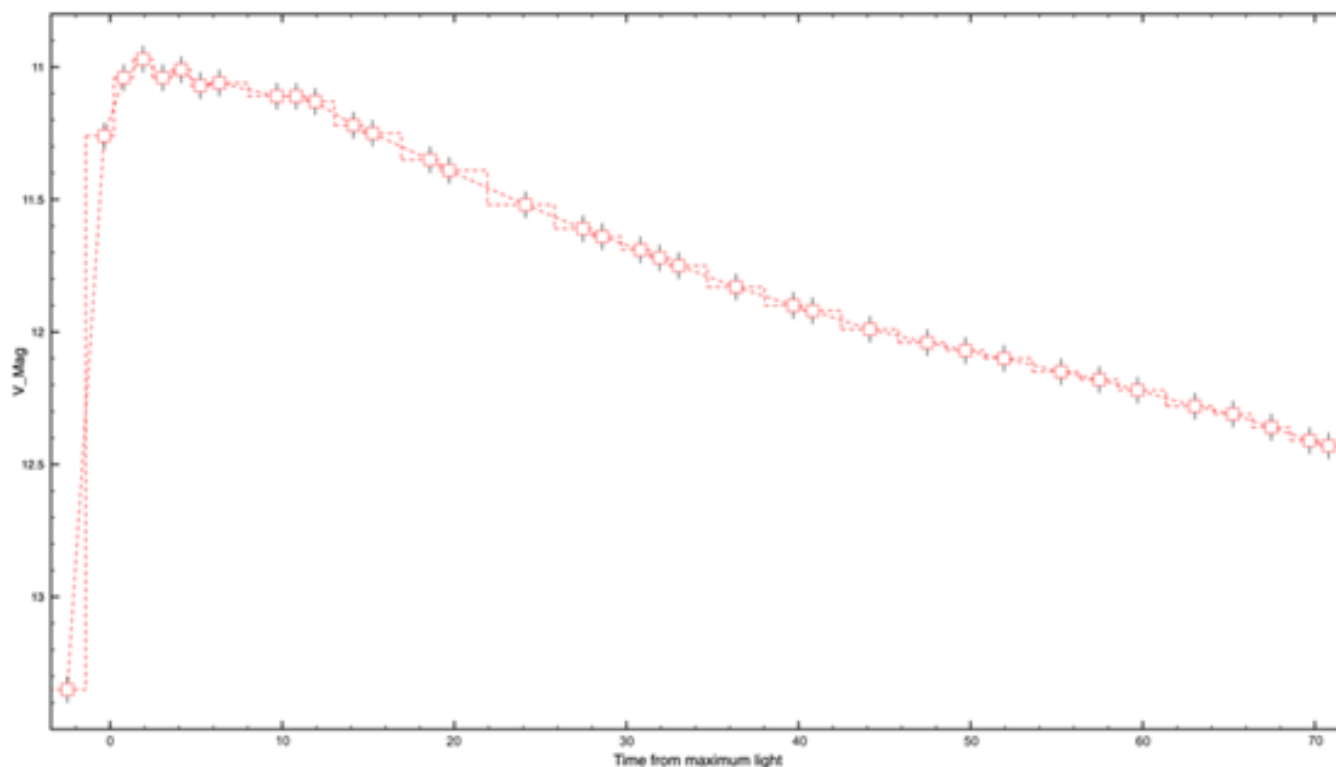


Figure 5 – The light curve obtained by the author for SN 2023ixf, as published in AstroNote 2023-

SJAA Member Astrophoto Gallery



Dates: July 29, 2023, Aug. 3, 2023, Aug. 4, 2023, Aug. 5, 2023, Aug. 6, 2023, Aug. 7, 2023, Aug. 8, 2023

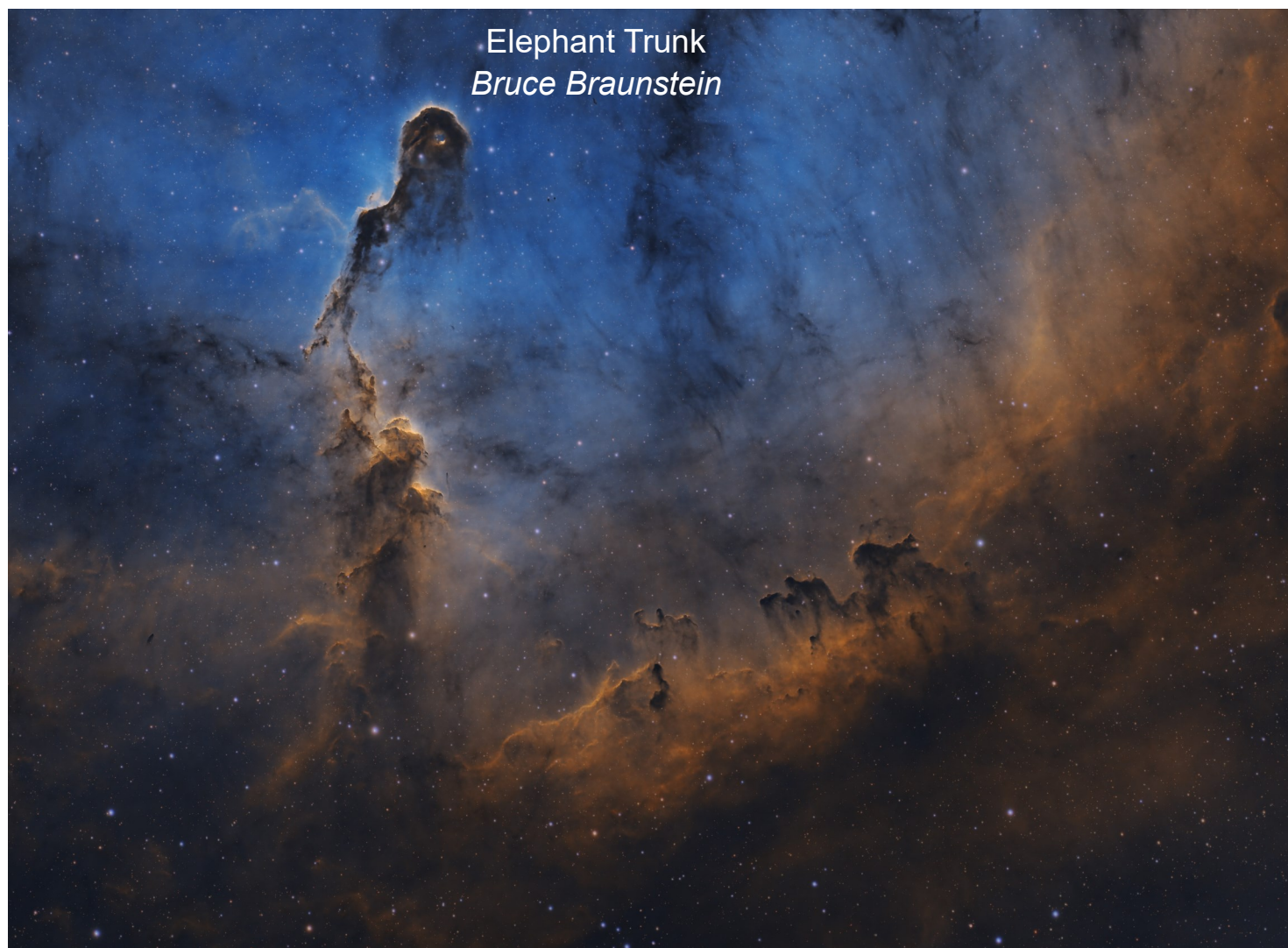
Location: Mountain View, CA

Imaging Telescope: Astro-Tech AT130EDT

Imaging Camera: ZWO ASI294MM Pro

Mount: iOptron iEQ45 Pro

Software: Adobe Photoshop, Cranfield / Payne Generalized Hyperbolic Stretching (GHS) Module/Script, Pleiades Astrophotography PixInsight, Russell Croman Astrophotography BlurXTerminator, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NI-NA)



Date: Aug. 4, 2023, Aug. 5, 2023, Aug. 6, 2023, Aug. 7, 2023

Location: Reno, NV

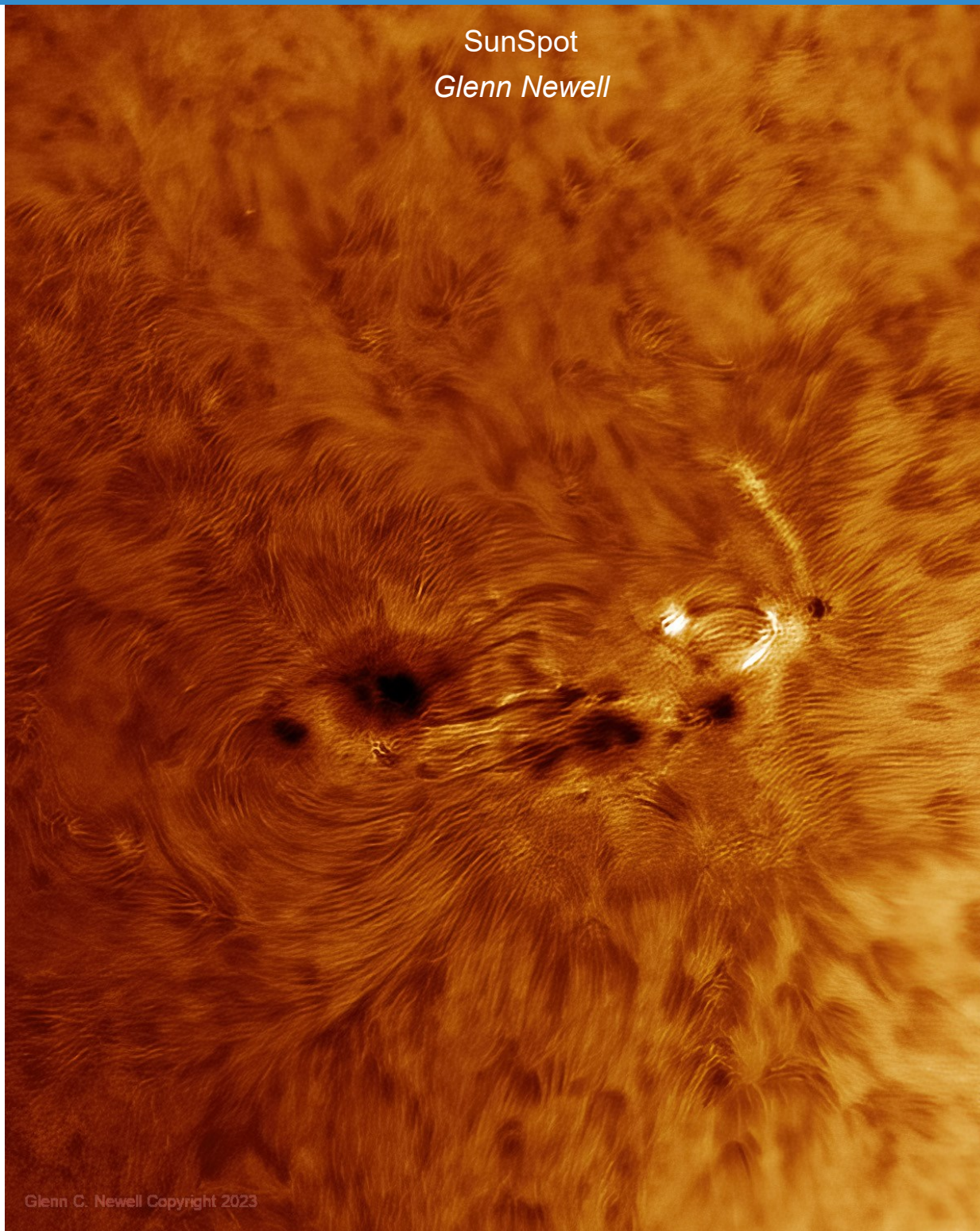
Telescope: Takahashi FSQ-106EDX4

Camera: QHYCCD QHY268 M

Mount: Software Bisque Paramount MX+ GEM

Software: Open PHD Guiding Project PHD2, Pleiades Astrophoto PixInsight, Russell Croman Astrophotography BlurX-Terminator, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator, Software Bisque TheSky, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)

SunSpot
Glenn Newell



Glenn C. Newell Copyright 2023

Dates: June 29, 2023

Location: Home, Union City, CA

Telescope: William Optics Megrez 80 ED II

Camera: ZWO ASI174MM COOL

Mount: Orion HDX110 Eq-G

Software: Sharpcap, AUTOSTAKKERT, IMPPG, Photoshop Topaz denoise and sharpen



Date: August 14, 2023

Location: San Jose, CA

Telescope: William Optics Redcat 71

Camera: ZWO ASI2600MC Pro

Mount: ZWO AM5

Software: Adobe Photoshop , Pleiades Astrophoto PixInsight

Practical Astronomy: Capturing A Solar Eclipse In Person

Credit: Jordan Makower



Annular Eclipse



Total Solar Eclipse



'Diamond Ring' Effect

There are approximately two solar and two lunar eclipses each year. Over 50% of the Earth can view a lunar eclipse, but a person must be in the right spot at the proper time to see any solar eclipse. The reason for this viewing problem is that the solar eclipse path is very narrow and, since about 70% of our planet is covered by water, a viewing site may be inaccessible.

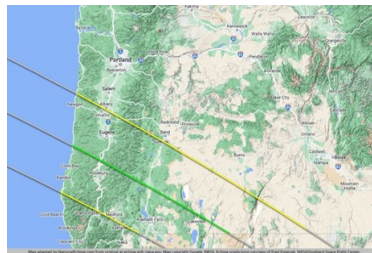
An Annular Eclipse will occur on October 14, 2023. Eight states, and areas in Central and South America will have 'front-row seats' to see it. On April 8, 2024, a Total Solar Eclipse will be visible (weather permitting) from eleven states, as well as portions of northeastern Canada, 18 areas in all.

Being at an eclipse site to experience the event is incomparable to anything else. During the eclipse, temperatures drop, reading a printed page is only possible with a flashlight, birds stop singing and, if near the ocean, wave actions and sounds are impressive. A person can't observe all this from a video or a still image! Showing up at a site without the proper equipment would be frustratingly disheartening.

Although *Sky & Telescope Magazine* and others will show the areas of visibility, timing of the eclipse, sun's altitude, probability of cloudiness, ..., by the time that information is printed, it might be too late to plan/find a safe viewing spot. Hotels, car rental agencies, and camping areas will have raised their prices. Even suppliers of optical/photographic gear will join the inflationary move. The best course of action is to find and secure a site, and purchase needed equipment before this happens.



Using binocular projection. Second camera lens is covered.



Annular Eclipse. Green line is centered path.



Path of Total Eclipse

Here's some of what a person might need. Special eclipse glasses, two digital cameras* with lens focal lengths greater than 400 mm (for large images), lens caps, **a special Solar Filter** (better than a 16.5 'stop' neutral density filter), tripod, remote camera 'trigger'/self-timer, consult Fred Espenak's website: eclipse.gsfc.nasa.gov to get exposure times/aperture/shutter speed for the separate stages of the eclipse, buy an electronic tracking mount if you can afford it (else you'll have to find a way to follow the Sun without using a viewfinder), binoculars ONLY USED to **project an image** on a screen, and extra batteries and memory cards. *The second camera is used to image the eclipsed Sun CAUTIOUSLY briefly without a filter!

The Unicorn Constellation

Credit: Jordan Makower

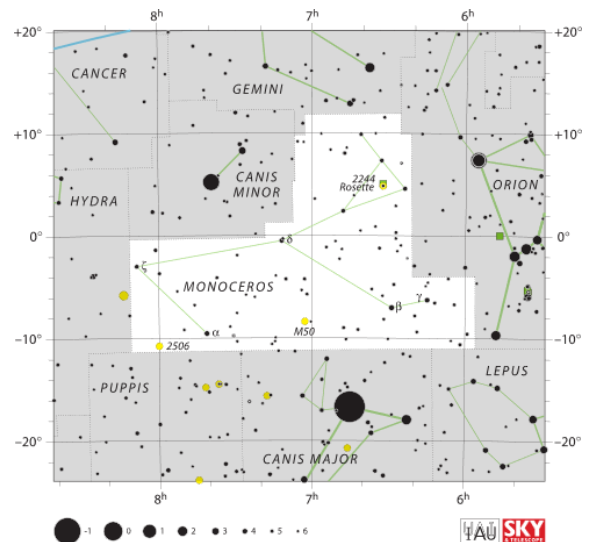
Our Association hosts **In Town Star Parties** usually once a month, on a Friday evening. Sometimes people discover these events during an outing to Houge Park, in San Jose. A few of our members are there, with their telescopes, to show people images of galaxies, nebulae, planets, the Moon, stars, ... I may be there, too, but without a scope, to assist with answering any questions about astronomy and to help people locate objects in the sky.

"I want to see a Unicorn Constellation", insisted a 5-year-old to her father, a new visitor to the park.



Sidney Hall – Urania's Mirror

Credit: Wikipedia.org



Monoceros

Credit: Wikipedia.org

Perhaps you know that the constellations are named after ancient legends/mythological beings/objects and, yes, there *is* a Unicorn Constellation, but there is no known legend to accompany it. As with other celestial objects, it bears no more resemblance to its namesake than Pegasus, Ursa Major, Cassiopeia, etc., do to theirs. Unicorns were drawn as early as 2600 BC and mentioned in natural history accounts by ancient Greeks (reference: Wikipedia).

Astronomers refer to this constellation as Monoceros, located to the East of Orion, Northeast of Canis Major, South of Canis Minor. It is rich in Messier Objects (M46, 47, 48), and bright nebulae, but has stars of low brightness, as shown in the chart above. It would best be found in an area away from city lights but, unlike Unicorns, persistent watching will be rewarded by finding this constellation.



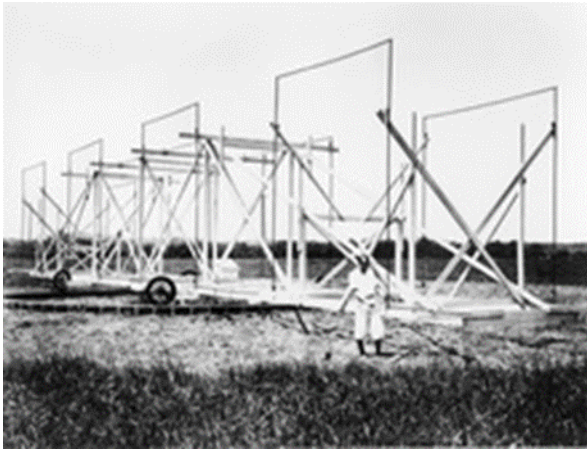
Kid Spot Jokes:

- ♦ **A photon walks into a hotel and the clerk asks if he needs help with luggage.**
- ♦ "No thanks!" says the photon. "I'm traveling light."
- ♦ **I was teaching my son astronomy last night and he chuckled when I pointed out the dog star.**
- ♦ Hey, I said this is Sirius!

Listen To Jupiter, Ganymede, The Sun, Galaxies

Credit: Jordan Makower

Karl Jansky, while working for Bell Laboratories and searching for the cause of steady radio static, discovered that the signal was coming from Sagittarius! He built a rotatable antenna designed to receive radio waves at 20.5 MHz, but a person can put together a simpler setup to receive astrometric energy with their own FM radio. Of course, the better the radio, and the longer the antenna, the more signals that will be received.



Jansky and his rotating directional radio antenna (early 1930s), the world's first radio telescope.

Credit: Wikipedia.org



FM Radio



FM Dipole

Credit: ebay.com

The Sun is the biggest radio source at most frequencies, but during times of minimal solar activity, waves from galaxies, Jupiter, Ganymede, geomagnetic storms, and other sources can be detected. Jansky, a physicist and radio engineer, recorded the signals he received with a pen-and-paper system, but we would hear it as a 'hiss' on an unused FM radio channel.

Signals from remnants of supernovas, neutron stars, star-forming/radio galaxies, extra-galactic pulses, and other objects can be detected with properly built/tuned radio antennas and receivers. Further research would give greater information about constructing such devices.

Several years ago, one of my students used her family's FM radio and a standard indoor FM dipole attached to a south-facing wall to record noontime solar 'static' and presented her project at 3 science conferences.

From the Board of Directors

Announcements

- ◇ Current SJAA Members can benefit from signing up for the "SJ Astronomy" group on Meetup (<https://www.meetup.com/sj-astronomy/>) as well as the "SJAA Observers" Google group (<https://groups.google.com/g/sjaa-observers>).

- ◇ Open Positions: Library

- ◇ "SJAA will be doing a partial eclipse event at our home base in Houge Park on the 14th. Volunteers are needed to welcome the crowd, share information, and staff various activities. Please contact Chris Gottbrath at vp@sjaa.net if you are interested in helping. We would particularly appreciate a few members with experience with hydrogen alpha or white light solar observing.

The eclipse starts at 8:06 am on October 14 and goes till 10:42 (with maximum eclipse at 9:20). The event details are still being worked out but it will likely start at 7:30 and go till 11 AM.

Note that this is a second event in addition to the event that Wolf and Nancy are organizing at KIPAC."

Board Meeting Excerpts

July 1, 2023

In attendance

Aroon Nair, Vini Carter, Peter Melhus, Ken Miura, Wolf Witt, Glen Newell, Chris Gottbrath, Swami Nigam

Absences: Rob Jaworski, Sukhada Palav

Guests: Nancy L, Michael M, Marianne D, Emi N

SJAA relationship with the City of San Jose

The board agreed to allow Marianne to invite personnel from the City of San Jose on July 14 to Astro 101 and ITSP.

Tilton Ranch Lunch

Peter made a motion to approve the reimbursement to Nancy as detailed above, Vini seconded. All approved, no abstentions or objections. Motion carried.

SJAA relationship with OSA

SJAA's relationship with OSA continues since the Tilton Ranch visit on June 24th. Nancy to continue driving this forward.

August 5, 2023

In attendance

Aroon Nair, Ken Miura Swami Nigam, Rob Jaworski, Glenn Newell, Chris Gottbrath

Absences: Peter Melhus, Sukhada Palav, Wolf Witt, Vini Carter

Guests: Emi Nikolov

Code of Conduct for SJAA leadership Team

SJAA formally adopted a Code of Conduct for its leadership team at the meeting of its Board of Directors on August 5, 2023. This Code of Conduct covers a variety of topics such as conflicts of interest, harassment and/or discrimination, confidentiality, and attendance and conduct at meetings. This code is applicable to all SJAA members in leadership position, such as Directors, Officers, Program Leaders, etc.

The process to get all leaders to agree to this Code of Conduct and sign it has been kicked off. The secretary will gather and keep all such attestations. The Code of Conduct has been posted on the SJAA website, to provide full visibility to the general public.

New OSA Portal

OSA is implementing a new volunteers portal in August. They will be asking us to log volunteer hours using this new portal for events that take place on OSA properties. This applies to SJAA members who are also OSA docents participating in programs such as Starry Nights, Binoculars Stargazing, Solar Observing, Imaging, as well as other Member Observing events. We will be receiving more information from them soon

Ongoing relationship SJAA and the

City of San Jose, Parks Department

AYS has shared a copy of the contract they signed with the city with us. The contract is valid until sometime in 2024. Rob & Marianne will go through the AYS contract in detail. Rob will share the contract with the board.

Minors policy and Code of Conduct

Aroon made a motion to approve the Code of Conduct policy as currently written up, Swami seconded. All approved, no abstentions or objections. Motion carried. The Code of Conduct policy has been formally adopted by SJAA as of August 5, 2023.

September 2, 2023

In attendance

Vini Carter, Ken Miura, Swami Nigam, Sukhada Palav, Peter Melhus, Glenn Newell, Aroon Nair, Chris Gottbrath, Wolf Witt

Absent: Rob Jaworski

Guests: Jonathan L, Joe T, Alex C, Larry Z, Nancy L, Michael M, Marianne D.

New Speaker Coordinator

Alex Colias is the new Speaker Coordinator replacing Sukhada in October!

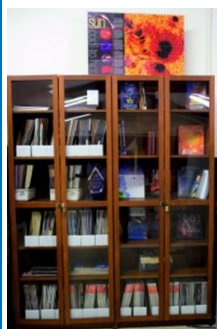
Annular Eclipse Event

SJAA has been invited by KIPAC (<https://kipac.stanford.edu/>) to contribute to their event that's expected to draw thousands of people. Wolf to send out an announcement about the KIPAC event to the members list.

Organizing "How to buy a telescope" workshop

Jonathan Lawton described the idea for planning a "How to buy a telescope" workshop with support from Joe Trdinich and Kanch. The board was very supportive of this idea and requested Jonathan and Joe to proceed with it. No formal vote was needed for this item.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STARt Program

The Quick STARt Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt program is now taking signups from members. In general will operate around ITSP in very small groups outside.

<http://www.sjaa.net/programs/quick-start/>

Astronomy 101

The Astronomy 101 session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular monthly session with a similar format, with the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided green laser tour

of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact Jon Anderson for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Director:	Aroon Nair
Vice-President:	Chris Gottbrath
Treasurer/Director:	Rob Jaworski
Secretary/Director:	Swami Nigam
Director:	Vini Carter
Director:	Wolf Witt
Director:	Ken Miura
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
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Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Astronomy 101:	Wolf Witt
In-Town Star Party:	Muditha Kanchana
Library:	Vacant
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Beth Johnson
Quick STARt	Jonathan Lawton
Donations:	Rich Gregor
Solar:	Wolf Witt
Starry Nights:	Joe Fragola
School Events:	Jon Anderson
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Alex Colias
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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